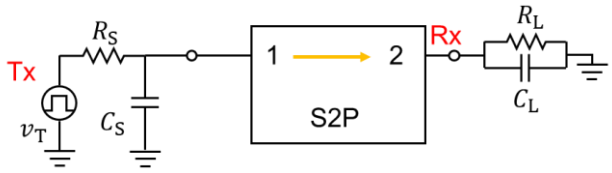
Homework: Jitter simulation and basic analysis

An S2P of a channel is provided to you, and the circuit is to be connected as below. The input v_T is 40 Gbps NRZ with parameters as shown.



$V_{pp} = 1$ V $\quad T_b = 25$ ps $\quad R_S = 20\ \Omega \quad R_L = 85\ \Omega$

$T_d = 0$ ps $\quad T_r = 5$ ps $\quad C_S = 0.3$ pF $\quad C_L = 0.3$ pF

Consider four cases:

- Case 1: Ideal Tx, no jitter. Namely, Rx only has data dependent jitter (DDJ).
- Case 2: Tx has sinusoidal jitter (SJ), with amplitude = 3 ps and frequency = 1 MHz.
- Case 3: Tx has random jitter (RJ), with zero mean and standard deviation σ = 1 ps.
- Case 4: Tx has both SJ and RJ, with the same parameters as Cases 2 and 3.

For each of the four cases, provide or compute the followings:

- Rx eye
- Tx jitter PDF (1-UI)
- Rx jitter PDF (1-UI)
- Tx jitter variance σ^2_{Tx} (a single value)
- Rx jitter variance σ^2_{Rx} (a single value)

Questions:

1. Does $\sigma^2_{\text{Tx, Case 2}} + \sigma^2_{\text{Tx, Case 3}} = \sigma^2_{\text{Tx, Case 4}}$? Should it or shouldn't it?
2. Does $\sigma^2_{\text{Rx, Case 1}} + \sigma^2_{\text{Tx, Case 2}} = \sigma^2_{\text{Rx, Case 2}}$? Should it or shouldn't it?
3. Does $\sigma^2_{\text{Rx, Case 1}} + \sigma^2_{\text{Tx, Case 3}} = \sigma^2_{\text{Rx, Case 3}}$? Should it or shouldn't it?
4. Does $\sigma^2_{\text{Rx, Case 1}} + \sigma^2_{\text{Tx, Case 4}} = \sigma^2_{\text{Tx, Case 4}}$? Should it or shouldn't it?

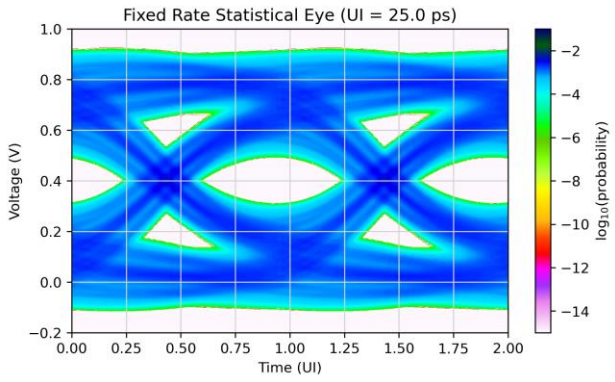
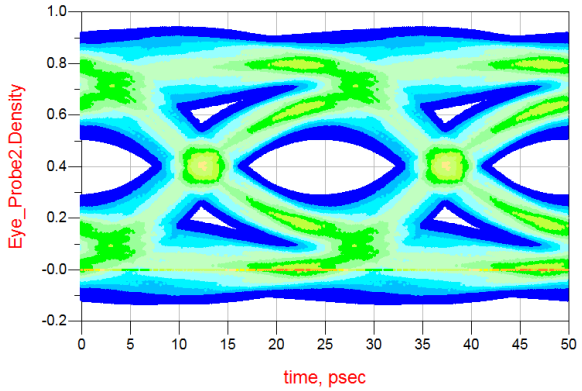
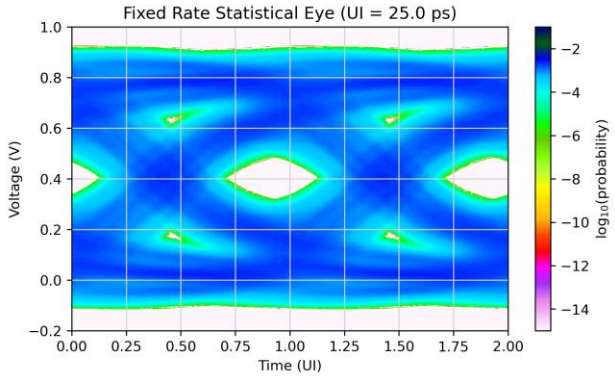
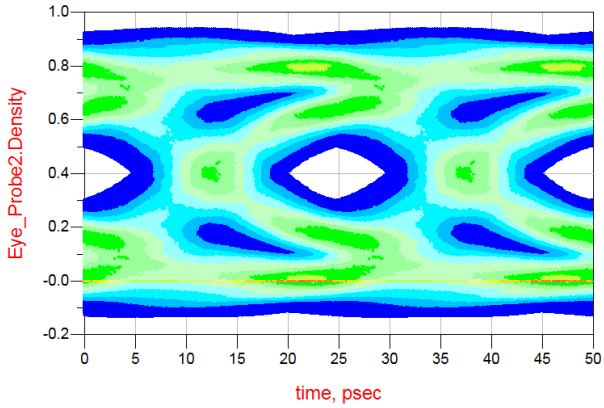
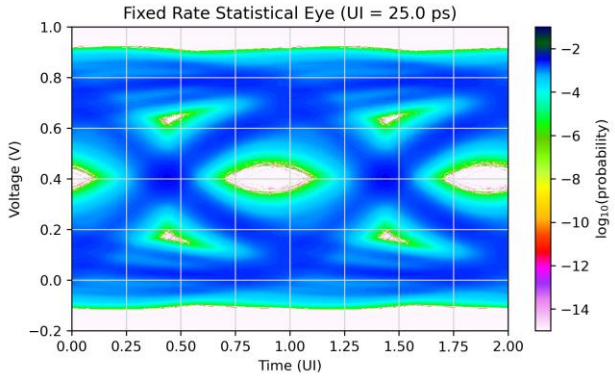
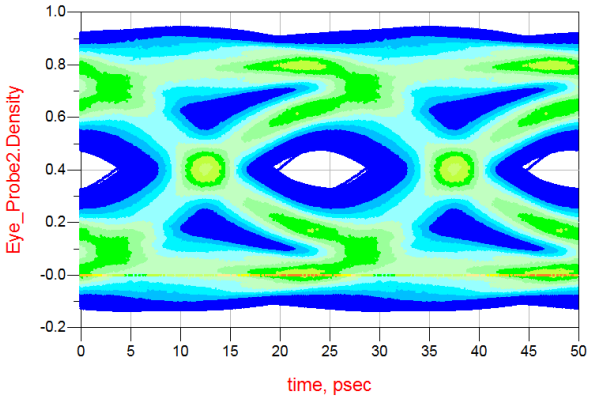
Notes:

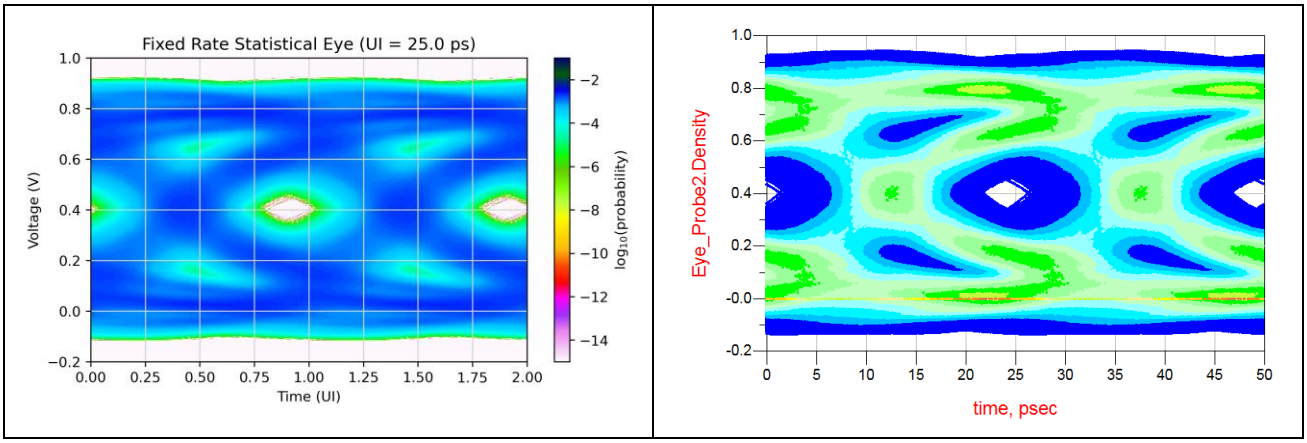
- To simulate jitter, the length of bit pattern should be 2^{20} or longer. This will give you some PDF data at regions where the probability is $2^{-20} \approx 10^{-6}$.
- For jitter analysis purpose, you should not add 10^{-15} to the array during constructing statistical eye, because it will affect the jitter distribution at low probabilities. Simply construct the eye as *it is*. You may add 10^{-15} or any small number to it when plotting, to avoid taking logarithm of 0.

Optional problem: Simulate the circuit in ADS ChannelSim to verify your results.

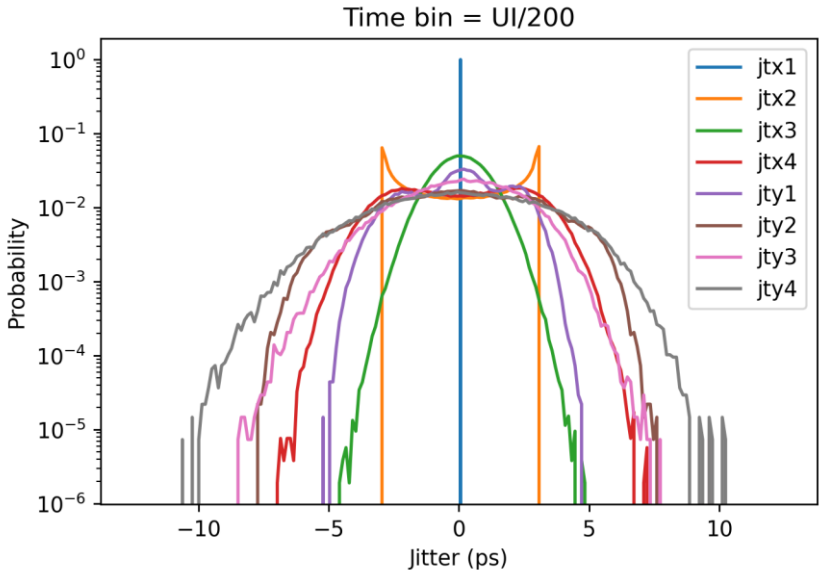
Appendix 1: Example answer

Nbp = 2^{20}

Python's eyes	ADS's eyes
Case 1: Tx no jitter	
	
Case 2: Tx PJ on	
	
Case 3: Tx RJ on	
	
Case 4: Tx PJ and RJ on	



Jitter distribution (this is the PMF of the specific bin size, not PDF)



Jitter rms at Tx

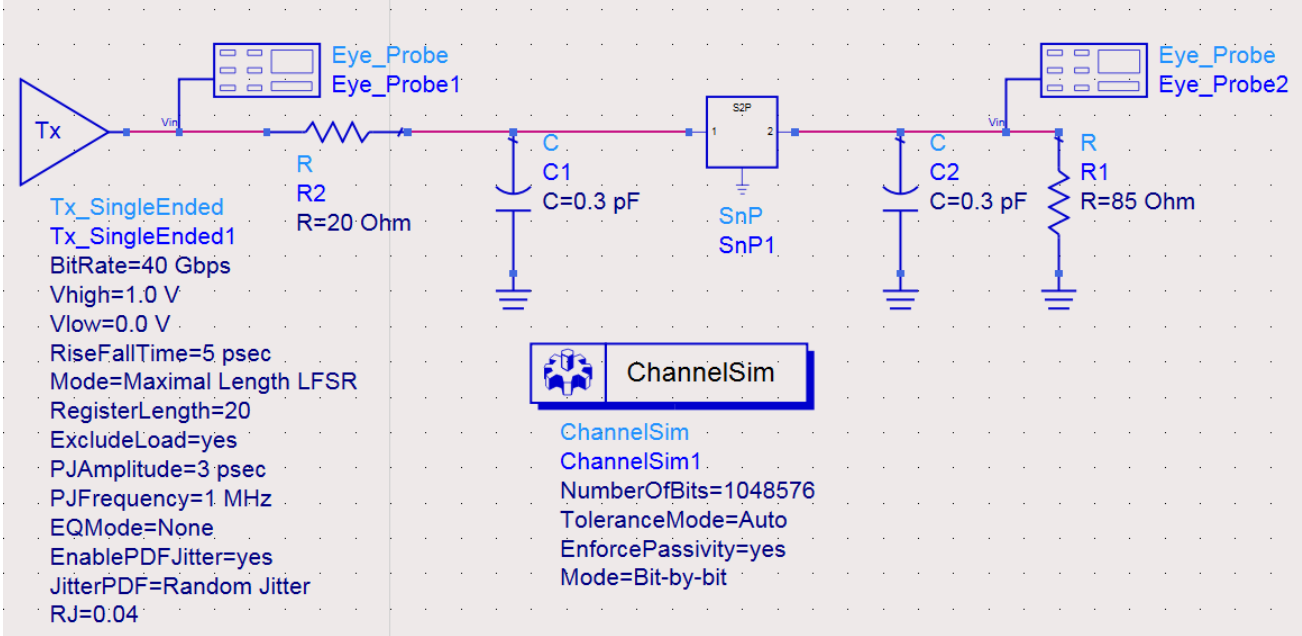
Jitter rms (ps)	Case 1 Tx	Case 2 Tx	Case 3 Tx	Case 4 Tx
Python	0	2.1247	1.0021	2.3459
ADS	0.062	2.3475	1.0074	2.3501
Theoretical	0	2.1213 (why?)	1	2.3452 (why?)

Jitter rms at Rx

Jitter rms (ps)	Case 1 Rx	Case 2 Rx	Case 3 Rx	Case 4 Rx
Python	1.7504	2.7540	2.0907	2.9747
ADS	1.8060	2.7799	2.0964	2.9737

Appendix 2: ADS setting

To simulate the eyes in ADS channelsim, connect the circuit as below.



Note that the unit of RJ is UI, so σ = 1 ps = 0.04 UI for 40 Gbps signal.

In the Eye_Probe setting, add the measurement of "JitterRMS".

